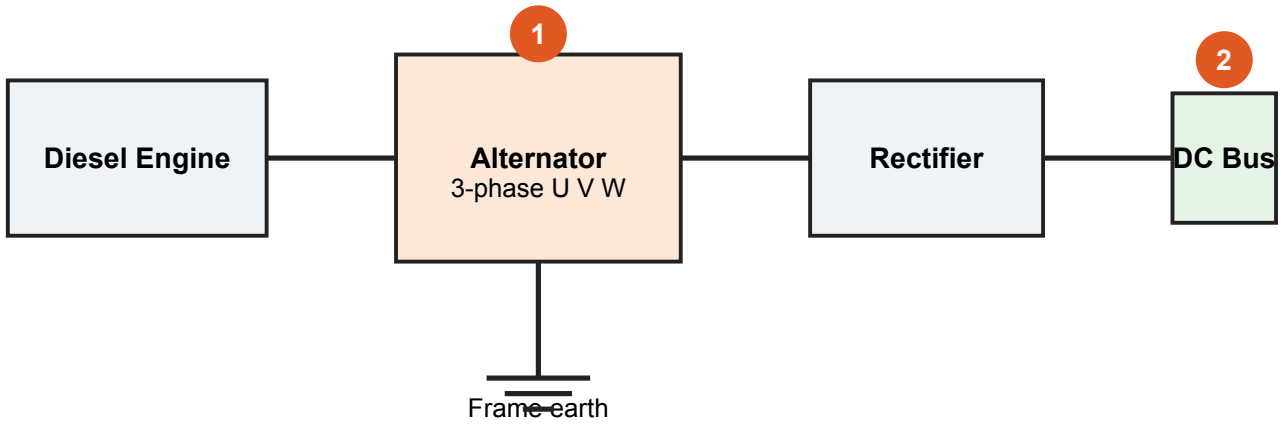
MEL-PRT-012418

Electrical System Inspection

MEL-SEED Haul Truck HT-02 - Alternator, Battery and Earthing

Maintenance Order	00012418 - HT02-SEED WO - Electrical System Inspection
Functional Location	MEL-SEED HT02 Alternator
Planner Group	MEL-SEED
Start Date	20.08.2026
Operation 0010	Electrical system inspection and measurements - 1.0 H

FIG 1 - Alternator and DC bus (HT-02)

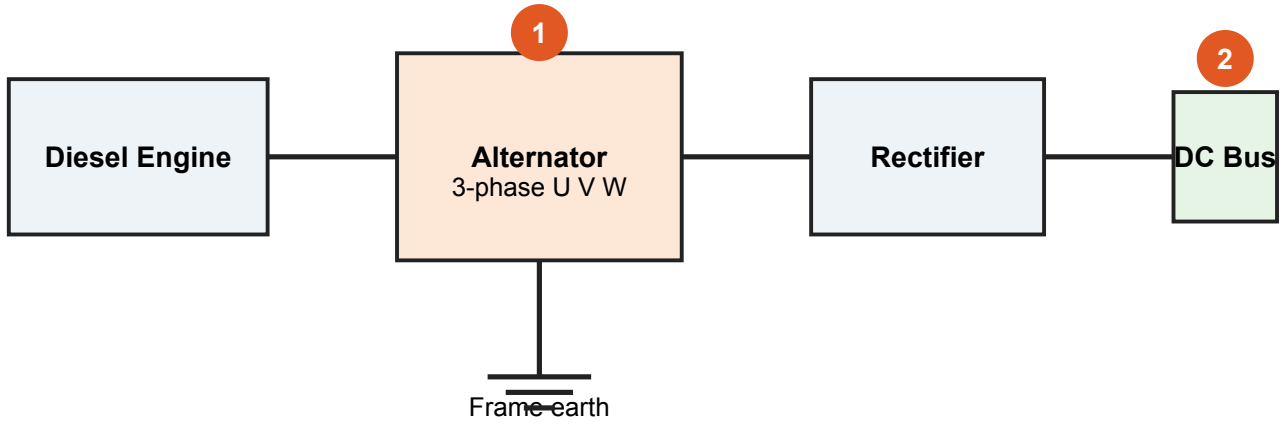


Purpose and Scope

AIM	The aim of this task is to inspect the HT-02 electrical system and record six electrical measurements in accordance with the maintenance plan.
INSTRUCTIONS	Electrical maintainer to complete the measures on pages 3 to 5 in order. Record each reading and tick OK, Attention or Scrap.
ISOLATIONS	Isolate and lock out battery isolator. Confirm zero energy before disconnecting any cable. Follow the site electrical isolation procedure.
PPE REQUIREMENTS	PPE as per site standards. Insulated gloves for all HV measures.
TOOLS & EQUIPMENT	Insulation tester (1000 V DC), multimeter, clamp meter, low-resistance ohmmeter, thermal imager.
SPECIAL NOTES	All maintainers shall tick all operations they complete on the service sheet and sign off. Completed sheets are to be handed to the relevant Supervisor.
Reference Documents	1. HT-02 Electrical Schematic Set - MEL-ESD-0042

Alternator Checks

FIG 1 - Alternator and DC bus (HT-02)



Measure 1 - Alternator Stator Insulation Resistance (U to Earth)

Isolate and disconnect the alternator output. Apply 1000 V DC between stator phase U and frame earth for 60 seconds. Record the stable value.

Acceptable: **≥ 100 MΩ** (see FIG 1, point 1)

Reading: _____ MΩ OK □ Attention □ Scrap □

Measure 2 - Alternator Output Voltage at DC Bus

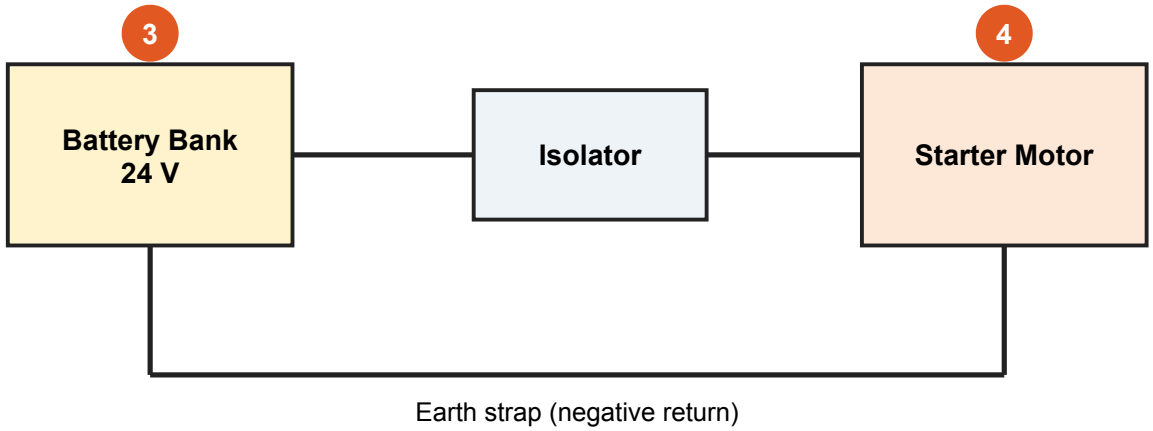
Run engine at rated idle with no load. Measure DC bus voltage at the bus terminals.

Acceptable: **590 - 610 V DC** (see FIG 1, point 2)

Reading: _____ V DC OK □ Attention □ Scrap □

Battery and Starting Circuit

FIG 2 - Battery and starting circuit (HT-02)



Measure 3 - Battery Bank Voltage at Rest

With the engine off for at least 30 minutes, measure across the battery bank terminals.

Acceptable: **24.0 - 25.5 V DC** (see FIG 2, point 3)

Reading: _____ V DC OK □ Attention □ Scrap □

Measure 4 - Starter Motor Current Draw

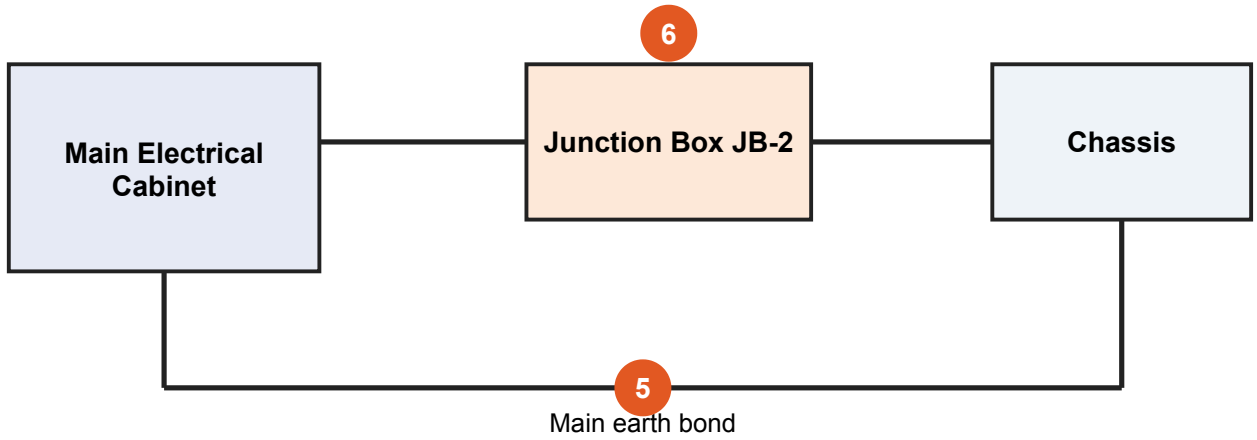
Clamp meter on the starter positive cable. Record peak current during one crank cycle of no more than 5 seconds.

Acceptable: **< 450 A** (see FIG 2, point 4)

Reading: _____ A OK □ Attention □ Scrap □

Cabinet and Earthing

FIG 3 - Cabinet, junction box and earth bond (HT-02)



Measure 5 - Main Earth Bond Continuity (Cabinet to Chassis)

Measure resistance between the main electrical cabinet earth bar and the chassis earth stud.

Acceptable: **< 10 mΩ** (see FIG 3, point 5)

Reading: _____ mΩ OK □ Attention □ Scrap □

Measure 6 - Junction Box JB-2 Terminal Temperature

Under normal load, scan junction box JB-2 terminals with the thermal imager. Record the hottest terminal.

Acceptable: **< 70 °C** (see FIG 3, point 6)

Reading: _____ °C OK □ Attention □ Scrap □

Comments:

_____

_____

_____

Name: _____ Sign: _____ Date: _____